Asymptotic notation

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The cost of an algorithm depends on the size n of its input. Asymptotic notation compares how two costs grow as n grows, ignoring constant factors and small inputs.

Upper, lower and tight bounds

Definition 0.1 (Big-O notation)**.** Let f and g be functions from the positive integers to $[0, \infty)$. We write $f(n) = O(g(n))$ if there are a constant $c > 0$ and an integer n_0 such that

$$f(n) \leq c\, g(n) \quad \text{for every } n \geq n_0.$$

That is, f grows no faster than a constant multiple of g once n is large enough. This is the definition of Arora and Barak (2009, sec. 0.3).

Definition 0.2 (Big-Omega notation)**.** For f and g as in Definition 0.1, we write $f(n) = \Omega(g(n))$ if there are a constant $c > 0$ and an integer n_0 such that

$$f(n) \geq c\, g(n) \quad \text{for every } n \geq n_0.$$

Dividing by c shows that this holds exactly when $g(n) = O(f(n))$, with constant $1/c$.

Definition 0.3 (Big-Theta notation)**.** For f and g as in Definition 0.1, we write $f(n) = \Theta(g(n))$ if both $f(n) = O(g(n))$ (Definition 0.1) and $f(n) = \Omega(g(n))$ (Definition 0.2). Then f and g grow at the same rate, up to constant factors.

Remark 0.1 (The equals sign is one-way)*.* $O(g(n))$ stands for a set of functions, and $f(n) = O(g(n))$ says that f belongs to it. So $n = O(n^2)$ is true, but $n^2 = O(n)$ is false, and the statement cannot be read from right to left. Cormen et al. (2009, sec. 3.1) reads the notation this way.

Dominance

Definition 0.4 (Little-o notation (dominates))**.** Let f and g be functions from the positive integers to $[0, \infty)$ with $g(n) > 0$ for every n. We write $f(n) = o(g(n))$ if, for every constant $\varepsilon > 0$, there is an integer n_0 such that

$$f(n) \leq \varepsilon\, g(n) \quad \text{for every } n \geq n_0.$$

That is, f is eventually smaller than any fixed fraction of g, which is the strict counterpart of Definition 0.1. In this case we also say that $g(n)$ **dominates** $f(n)$.

Example 0.1.

A cubic dominates a quadratic
Let $f(n) = n^2$ and $g(n) = n^3$. Let $\varepsilon > 0$. Choose an integer $n_0 \geq 1/\varepsilon$. For every $n \geq n_0$ we have $1/n \leq 1/n_0 \leq \varepsilon$, so

$$f(n) = n^2 = \frac{1}{n}\, n^3 \leq \varepsilon\, n^3 = \varepsilon\, g(n).$$

Hence $n^2 = o(n^3)$ by Definition 0.4, so n^3 dominates n^2.

Examples

Example 0.2 (A quadratic is $O(n^2)$ but not $O(n)$)**.** Let $f(n) = 3n^2 + 5n$.
For $n \geq 5$, we have $5n \leq n \cdot n = n^2$, so

$$\begin{aligned} f(n) &= 3n^2 + 5n \\ &\leq 3n^2 + n^2 \\ &= 4n^2. \end{aligned}$$

So $f(n) = O(n^2)$, with $c = 4$ and $n_0 = 5$ in Definition 0.1. Since $f(n) \geq 3n^2$ for every n, also $f(n) = \Omega(n^2)$ (Definition 0.2), and so $f(n) = \Theta(n^2)$ (Definition 0.3).
But $f(n)$ is not $O(n)$. If it were, there would be $c > 0$ and n_0 with $3n^2 + 5n \leq c\, n$ for every $n \geq n_0$. Dividing both sides by $n > 0$ gives $3n + 5 \leq c$ for every $n \geq n_0$, which fails for any n larger than both n_0 and $(c - 5)/3$.

Example 0.3 (Counting the operations in an inner product)**.** The inner product of two vectors $\underset{\sim}{x}, \underset{\sim}{z} \in \mathbb{R}^D$ is

$$\underset{\sim}{x}^{\top}\underset{\sim}{z} = x_1 z_1 + x_2 z_2 + \cdots + x_D z_D.$$

Computing it takes D multiplications, one for each product $x_d z_d$, and $D-1$ additions to sum the D products: $2D-1$ arithmetic operations in all. Since $2D-1 \leq 2D$ for every $D \geq 1$, the cost is $O(D)$, with D in the role of n, $c = 2$ and $n_0 = 1$ in Definition 0.1. Since $2D-1 \geq D$ for every $D \geq 1$, it is also $\Omega(D)$ (Definition 0.2), so it is $\Theta(D)$ (Definition 0.3). When a machine learning text says an inner product "costs $O(D)$", it means this count of operations, up to a constant factor.

References

Arora, Sanjeev, and Boaz Barak. 2009. *Computational Complexity: A Modern Approach*. Cambridge University Press. https://doi.org/10.1017/CBO9780511804090.

Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. 2009. *Introduction to Algorithms*. 3rd ed. MIT Press.